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Silent fish surveys: bubble-free diving highlights inaccuracies associated with SCUBA-based surveys in heavily fished areas

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Summary

1. Underwater visual census (UVC) using SCUBA is a commonly used method for assessing reef fish communities. Evidence suggests, however, that fish avoid divers due to the sound of bubbles produced by open-circuit SCUBA, and avoidance behaviour is more pronounced as fishing pressure increases. Despite the potential for producing biased counts and conclusions, these behavioural effects have rarely been quantified, especially when assessing the effectiveness of marine protected areas (MPAs).

2. To test the magnitude of avoidance behaviour, we surveyed fish populations inside and outside two MPAs in Guam, using two diving systems: standard open-circuit (OC) SCUBA and a closed-circuit rebreather (CCR) that produces no bubbles. Data were collected using a diver-operated stereo-video system (stereo-DOV), which provided counts of relative abundance, measures of fish length and the minimum approach distance of the diver to a fish.

3. Inside MPAs, fish surveys conducted with CCR recorded similar community metrics to fish surveys conducted with conventional OC SCUBA. In contrast, outside the MPAs, the bubble-free diving system recorded 48% more species and up to 260% greater fish abundance. These differences reflected the ability of a diver wearing the silent CCR unit to sample the larger, most heavily targeted species that are shy of divers in fished areas. This difference was also large enough to change some results from 'reject' to 'accept' the null hypothesis of 'no significant differences exist between fished and protected areas'.

4. The use of CCR for fish surveys clearly minimizes behavioural biases associated with fish avoiding open-circuit SCUBA divers. We recommend the use of this bubble-free diving system for surveys assessing reef fish populations, especially in areas where fish are heavily targeted by spearfishing. If fish behaviour is not accounted for, surveys using SCUBA could result in erroneous conclusions when comparing fished and protected areas. While the behaviour of fish towards divers is rarely mentioned in conclusions from studies using UVC, it is an important source of bias that should be acknowledged and minimized where possible.

Key-words: closed-circuit rebreather, marine protected area, underwater visual census, fish behaviour, minimum approach distance, flight initiation distance, spearfishing, observer presence, survey bias, stereo-video

Introduction

Ecological field studies require survey techniques that can accurately and consistently measure the relative abundance, size and distribution of organisms. Observational census techniques have long been used for non-cryptic mobile organisms such as large mammals, birds and fish (Thresher & Gunn 1986; Pollock & Kendall 1987; Thompson 2002). However, the presence of an observer has been shown to have an effect on animal behaviour with semaphore crabs (MacFarlane & King 2002), vervet monkeys (McDougall 2012) and burrowing owls (Manning & Kaler 2011), all changing their behaviour as a result of an observer being present. In the marine realm, Stoner *et al.* (2008) observed that their largest bias when sampling fishes with a remote operated vehicle (ROV) was associated with fish attracted to, or avoiding the ROV outside the field of view. They concluded that the distance at which fish behaviour changed was unknown and that it was important to understand and quantify the effects of observational bias so that it could be incorporated into the survey design.

On coral reefs, resident fish populations are typically surveyed by SCUBA divers using underwater visual census (UVC) methods (e.g. Thresher & Gunn 1986; Samoilys & Carlos 2000). It is one of the most popular methods as it does not require a lot of equipment and is therefore a cost-effective means of surveying reef fish in shallow waters. Yet SCUBA diving is recognized as an intrusive activity that can influence

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fish behaviour thereby distorting any measure of their distribution and abundance (Lobel 2001; Schmidt & Gassner 2006; Dickens *et al.* 2011). Early research using underwater sonar found that fish could detect divers well before the divers saw the fish and that this could have significant impacts on population estimates (Chapman *et al.* 1974; Chapman & Atkinson 1986; Stanley & Wilson 1995). Despite these recognized biases, only ~30% of 100 UVC studies conducted between 1999 and 2009 acknowledge any potential effect that the presence of a SCUBA diver might have on fish behaviour and population estimates (Dickens *et al.* 2011). Although some authors have stated that changes in fish behaviour could impact UVC results, the interaction between observers and fish is rarely quantified.

Biases due to altered fish behaviour are assumed not to be an issue for fish surveys if the biases are consistent across the spatial scales investigated. However, recent studies suggest that fish behaviour may change at small spatial scales, particularly inside and outside areas closed to fishing. For example, targeted fish species will consistently flee from an observer at a greater distance in fished areas compared to protected areas (Gotanda, Turgeon & Kramer 2009; Feary *et al.* 2011; Januchowski-Hartley *et al.* 2013). Moreover, avoidance behaviour of fish towards SCUBA divers becomes greater with increasing disturbance levels, especially in areas targeted by spearfishers (Kulbicki 1998; Guidetti, Vierucci & Bussotti 2008). Whereas in protected areas, fish can become accustomed or attracted to SCUBA divers when presented with food (Chapman *et al.* 1974; Cole 1994). It is therefore imperative we recognize that fish behaviour towards SCUBA divers might influence population estimates of fishes in areas of different fishing pressure.

While a comprehensive assessment of the stimuli associated with fish avoidance behaviour has not been done, sound has been implicated as a significant contributing factor (Chapman *et al.* 1974; Lobel 2005; Cole *et al.* 2007). The noise level of bubbles emitted from open-circuit (OC) SCUBA diving is in the low frequency range (<400 Hz), where fish hearing is most sensitive (Popper 2003; Lobel 2005; Radford *et al.* 2005). Additionally, bubbles create a visual disturbance barrier (Sharpe & Dill 1997) which adds to the overall impact associated with SCUBA-based UVC methodology. Removing bubbles as a source of disturbance should therefore create a more serene environment within which to conduct UVC and result in data that more accurately reflects the natural state of the fish assemblages being surveyed.

Use of a closed-circuit rebreather (CCR) does not produce bubbles if operated at a constant depth, hence providing a suitable tool for estimating the effects of SCUBA-derived bubbles on the quality of data collected by diver surveys. While the CCR diving technique has been recommended for behavioural observations of marine life since the 1970s (Collette & Earle 1972; Hanlon *et al.* 1982; Lobel 2001), we found no published studies comparing OC SCUBA and CCR for quantitative fish surveys (but see Cole *et al.* 2007 for the use of semi-closed-circuit rebreather). Our study was designed to quantify bubble effects through comparisons of fish community metrics collected by the two different diving techniques. By minimizing fish avoidance behaviours, we expected that a diver using CCR could approach closer and hence record a greater diversity, abundance and biomass of fish than divers using OC SCUBA. We included an interaction effect with marine protected areas (MPAs) to specifically investigate the hypothesis that fishing pressure creates behavioural changes in fish which may increase the differences between diving techniques.

Materials and methods

STUDY AREA

Fish surveys were conducted on the coral reefs of Guam, the largest and southernmost of the Mariana Islands in Micronesia (Fig. S1). Guam has a network of MPAs that provide reference areas for the impact of fishing on the fringing coral reefs. Enforced in 2001, the protected areas are not strictly no-take areas (allowing limited fishing from shore and trolling for pelagic species), but they do prohibit spearfishing within their boundaries (Burdick *et al.* 2008).

Two locations were chosen, each of which included an MPA and adjacent area of similar size and habitat (Fig. S1). The Guam West location included the Tumon Bay Marine Preserve which covers 4·52 km^2 (Burdick *et al.* 2008). Due to the proximity to the main population centre, recreational SCUBA divers frequently visit the MPA and enforcement is high. The other location, Guam South, which includes the Achang Marine Preserve (4·85 km^2), is located away from the main population centre and only rarely visited by recreational divers.

SURVEY EQUIPMENT

Diving equipment

Surveys were conducted using a CCR (Inspiration Classic; Ambient Pressure Diving Ltd, Cornwall, UK) and standard open-circuit (OC) SCUBA equipment. When using a CCR, exhaled breath is recirculated through a chemical carbon dioxide absorbent material and oxygen is only added to replace that lost through metabolism (Sieber & Pyle 2010). Hence, CCRs do not produce bubbles if used at a constant depth. This is in contrast to the commonly used OC diving system where each breath of gas is delivered from a compressed air cylinder and expelled into the water through a demand-valve regulator.

Diver-operated stereo-video system

A diver-operated stereo-video system (stereo-DOV) was used to sample reef fish (Watson *et al.* 2010). The stereo configuration of the video cameras (Sony HDR-CX12) allows accurate measurements of fish length, distance between the video camera and the focal fish, and allows the sampling area to be standardized (Shortis, Harvey & Abdo 2009). The configuration, accuracy and precision of the camera system used in this study have been presented in detail by Harvey *et al.* (2010).

DATA COLLECTION

Diving surveys

At each location (Guam West and Guam South), three sites were sampled inside and outside the MPAs using CCR and OC diving techniques. Fish were surveyed along five replicate transects 50 m

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long by 5 m wide (250 m^2 of reef) at each site. The transect dimension and level of replication was chosen as it is commonly used for quantifying coral reef fishes (Halford & Thompson 1994; Samoilys & Carlos 2000). The length of transect was calculated and marked out using a modified hip chain counter (Chainman Industries) which leaves a thin line of biodegradable cotton on the reef. The use of cotton avoids the presence of fibreglass tape measures which may also affect the fish community during repeat surveys (Dickens *et al.* 2011). The same transects were sampled by each diving technique with the starting order of sites and techniques randomised. A timeout period of 60 min was taken between repeat sampling of the same area to reduce disturbance on the fish community between surveys. A total of 120 transects were completed (60 from each of the CCR and OC systems) in June 2011 (Guam West) and October 2011 (Guam South).

Video analysis

Stereo-video footage was analysed using the software EventMeasure-Stereo (SeaGIS Pty Ltd) which logs and records the stereo coordinates of each measurement. We identified and measured the length of all non-cryptic fish species that are regularly caught in Guam's coral reef fishery and the conspicuous, non-target families Chaetodontidae, Pomacanthidae and Zanclidae (Table S1). Fish lengths were measured at the closest point to the diver, but if the diver approached the fish at a closer distance where the length measurement was not possible (due to the angle of the fish or the camera view being obstructed), a '3D point' was recorded and used as the distance for the measured fish. We did not include fish that were outside the sampling area (2·5 m either side of the centre of the stereo-DOV) or at a distance greater than 10 m ahead of the cameras. For each fish, we also recorded the conspecific group size, life phase that is initial phase (IP) or terminal phase (TP) of sexually dimorphic species, or if the fish was juvenile (usually <100 mm for targeted species).

Minimum approach distance

For every fish sampled, the distance between the diver and the fish at its closest point was termed 'minimum approach distance'. This was used as a measure of fish behaviour towards the diver and is similar to the measure of flight initiation distance used by other authors studying coral reef fish (Gotanda, Turgeon & Kramer 2009; Januchowski-Hartley *et al.* 2011). In contrast to flight initiation distance, distance was recorded for every fish on the transect, even if the fish did not flee and moved away during what appeared to be normal swimming activity.

Biomass calculation

Biomass (grams) was calculated from length measurements (mm fork length) using length–weight relationships. Length–weight regression values were derived from fishery catch data in Guam, but when not available (i.e. for non-target species), these values were sourced from FishBase (Froese & Pauly 2013). Reef sharks (Carcharhinidae) were excluded from biomass analysis as their rare occurrence and large size would disproportionally affect biomass patterns when present. For individual fish that were not measured (e.g. being obscured from one of the camera views), we calculated biomass using an average length for that species from the site where it occurred.

Grouping of fish

For data analysis, fish species were placed into four groups: fished, target, non-target and high value (Table S1). FISHED species included all non-cryptic fish species that are regularly caught in Guam's coral reef fishery (excluding juveniles). Parrotfish; scarines (family: Labridae, tribe: Scarinae), and the surgeonfish, unicornfish and tangs; acanthurids (family: Acanthuridae), were analysed separately as they are the main family groups caught by spearfishing in Guam (Lindfield, McIlwain & Harvey 2014). TARGET species from these two groups were selected by excluding species that were not targeted by spearfishers in Guam (desirability score <1) according to Bejarano *et al.* (2013). The larger brightly coloured terminal phase (TP) of the parrotfishes, *Chlorurus spilurus* (formally *sordidus*), *Scarus globiceps* and *Scarus psittacus*, was classified as targeted, whereas the smaller initial phase (IP) was not. We further classified species as HIGH VALUE if desirability scores were >3 (Bejarano *et al.* 2013) or identified as significant contributors to reef fish catch in Guam (Houk *et al.* 2012), in addition to targeted species from families Lethrinidae, Epinephelidae, Lutjanidae and Carcharhinidae.

STATISTICAL ANALYSIS

As our experimental design consisted of sites nested within MPA status and different diving techniques repeated on the same transects, mixed effects model techniques were used for the analysis of univariate response variables (Bolker *et al.* 2009; Zuur *et al.* 2009). Mixed effects models can account for dependencies within hierarchical groups through the introduction of random effects and are well suited for repeated measures designs (Pinheiro & Bates 2000). All statistical modelling was conducted with the lme4 package (Bates *et al.* 2014) in the R language and environment (R Core Team 2014). Graphical plots were created with the ggplot2 package (Wickham 2009).

To test for differences in count data (abundance, biomass and species richness), we used generalised linear mixed models (GLMMs) on the untransformed data by fitting a Poisson distribution (log-link) using Laplace approximation (Bolker *et al.* 2009). The optimal random effects structure was determined (using log likelihood ratio tests) after fitting the full model with varying random effect structures (Zuur *et al.* 2009). This process consistently resulted in a random intercept for *Site* and random slope of *Technique* within *Transects* with a correlated intercept. Including transects into the design (with different random effects structure between the different diving methods) also assured the model was not overdispersed. We then ran the full model with the categorical variables *Status* (MPA or fished) and *Technique* (OC or CCR) and their interaction as fixed effects and checked model validation results (Bolker *et al.* 2009; Zuur *et al.* 2009).

Minimum approach distance of fishery-targeted acanthurids and scarines and the non-target chaetodontids was analysed using linear mixed models (LMMs) with Gaussian distribution. We included fixed effect covariables, group size and length of fish as they are known to influence flight initiation distance for reef fish (Gotanda, Turgeon & Kramer 2009; Januchowski-Hartley *et al.* 2011). Group size showed large non-normal variation and was therefore log-transformed. Optimal random effect structure was typically similar to GLMM analyses but sometimes required a random group intercept for *Transects*. For the full model, we tested the significance of each fixed effect using log likelihood ratio tests and non-significant effects were removed before re-running the analysis to obtain the optimal mixed model (Zuur *et al.* 2009). We present final model results with p-value significance calculated from

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t-tests using the lmerTest package (Kuznetsova, Brockhoff & Christensen 2014).

To test for differences in assemblage composition of all fish species recorded, we conducted permutational multivariate analysis of variance (PERMANOVA) using PRIMER v6 (Anderson, Gorley & Clarke 2008). Analysis followed the same statistical design as the mixed models with *Sites* (nested within *Status*) and *Transects* (nested within *Sites*). Our experimental design was balanced, and therefore, PERMANOVA will be robust to heterogeneity when testing for differences in the location of centroids among the groups (Anderson & Walsh 2013). A Bray–Curtis dissimilarity matrix was created between every pair of observations to focus on the change in the composition of species identities. Prior to analysis, biomass data were $\ln(x + 1)$-transformed to remove large differences in scale among the original variables (Anderson & Willis 2003). Principal coordinate analysis (PCO) was then used to examine the relationships between the samples through a direct projection defined by dissimilarities (Anderson & Willis 2003). Spearman rank correlations of $|r| > 0{\cdot}45$ were used to show relationships between individual species and the PCO axes.

Results

FISHED SPECIES

Abundance

Surveys conducted using OC SCUBA recorded 1·5–2·6 times the mean abundance of fished species within the MPA compared to fished areas at Guam South and Guam West, respectively (Fig. 1a). In contrast, surveys using CCR recorded similar numbers of fish between MPA and fished areas at both locations with no overall effect of MPA status in the GLMM analysis and a significant interaction term of Status x Technique (Table S2). Furthermore, the magnitude of difference between the two diving techniques, particularly at Guam West, revealed that CCR recorded 260% greater average fish abundance than OC SCUBA in fished areas (56·3 vs. 21·7 fish per transect), whereas numbers were similar within the MPA (Table S2, Fig. 1a).

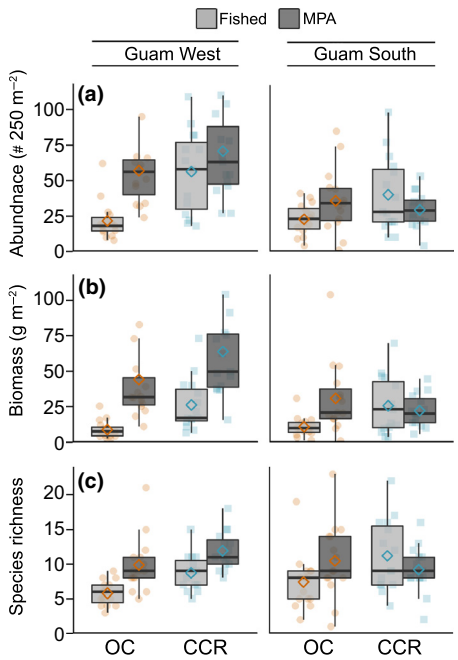

Fig. 1. Boxplots comparing (a) abundance, (b) biomass and (c) species richness of fished species between MPA and fished areas for each diving technique. Mean values are displayed as diamonds for each contrast, and values from replicate transects are depicted in colour for diving techniques (orange: OC, blue: CCR).

Biomass

Surveys conducted using OC SCUBA recorded over five times the mean of biomass of fished species inside the MPA compared to fished areas at Guam West and 2·9 times the mean biomass at Guam South (Fig. 1b). CCR surveys also recorded greater biomass within the MPA at Guam West, but differences were of a smaller magnitude (2·4 times) and no MPA effect was evident at Guam South (Table S2, Fig. 1b). In fished areas, less biomass of fish was recorded with OC, averaging only 30–40% of the biomass observed with diving with CCR (Fig. 1b). However, in protected areas, these differences were not significant (Table S2, Fig 1b).

Species richness

The number of fished species recorded between diving techniques was similar within MPAs (CCR = 52 species, OC = 53 species). In contrast, at fished sites, CCR recorded 48% more species than OC (49 vs. 33). At Guam West, species richness values were greater for CCR surveys than OC at both fished and protected sites, and both diving techniques recorded higher species richness within the MPA compared to fished areas (Table S2, Fig. 1c). Whereas at Guam South, lower species richness was observed only when surveys conducted with OC in fished areas (Table S2, Fig. 1c).

TARGET AND NON-TARGET GROUPS

Abundance

Targeted acanthurid (surgeonfish, unicornfish and tang) abundance was very low in fished areas at Guam West, irrespective of diving technique. Whereas at Guam South there was little difference between fished and protected areas, but less fish were recorded with OC diving compared to CCR (Table S3, Fig. 2a). Likewise, the abundance of targeted scarines (parrotfish) was lower when surveyed with OC compared to CCR at fished locations but not within MPAs (Table S3, Fig 2b). At the Guam West location, differences between MPA and fished areas were most pronounced when surveys were conducted with OC SCUBA. In contrast, the non-target chaetodontids (butterflyfish) showed no significant differences in biomass due to MPA status or between diving techniques (Table S3, Fig 2c).

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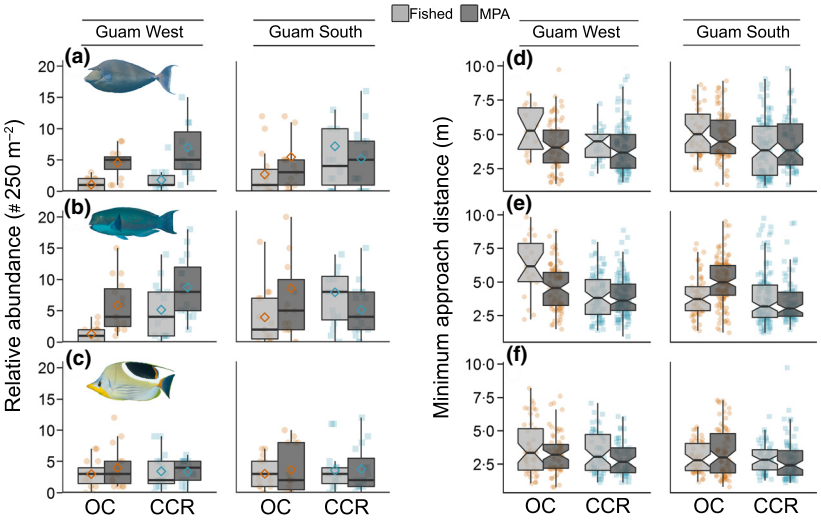

Fig. 2. Boxplots comparing the abundance (left panel) and minimum approach distance (right panel) between MPA and fished areas for each diving technique. Fish groups are compared along rows for (a,d) acanthurids, (b,e) scarines and (c,f) chaetodontids. Each replicate value is depicted in colour for diving techniques (orange: OC, blue: CCR).

Minimum approach distance

With the exception of scarines at Guam South, OC diving in fished areas had the greatest average minimum approach distance for targeted fish groups (Table 1, Fig. 2d,e). This greater minimum approach distance indicates that fish avoid the observer by staying further away which also corresponds to the lowest biomass values (Fig. 2a,b). At the fished areas of Guam West, a diver using OC SCUBA would not approach these targeted species closer than 5 m on average (Fig. 2d,e). The use of CCR at the same sites allowed the diver to approach acanthurids on average 1 m closer than OC diving (5·3 vs. 4·3 m) and 2 m closer for scarines (6·1 vs. 4·0 m). Group size and fish length were also important explanatory variables (Table 1). Increasing length was positively associated with increasing minimum approach distance (i.e. large fish would stay further from the diver), whereas a larger group size would generally allow divers to approach the fish closer. For the non-targeted chaetodontids, distance of approach was not affected by diving technique or MPA status (Table 1, Fig. 2f).

ASSEMBLAGE STRUCTURE

At Guam West, fish assemblages differed between diving techniques at fished sites but not at MPA sites (significant interaction of Status × Technique; Table S4). A similar trend was also apparent at Guam South; however, it was not significant at the $P < 0{\cdot}05$ level ($P = 0{\cdot}077$). Assemblage differences are visualized with a PCO ordination (Fig. 3), which illustrated separation of the OC fished replicates from the CCR fished replicates, the latter being more similar to samples from the MPA (Fig. 3). At Guam West, the separation between diving techniques at fished sites was primarily correlated with the biomass of small-bodied herbivorous species on the first PCO axis,

Table 1. Summary table of the optimal linear mixed models (LMMs) for minimum approach distance (m)

	Guam West				Guam South			
Effect	Estimate	SE	*t* value	Pr(>|t|)	Estimate	SE	*t* value	Pr(>|t|)
Acanthurids								
(Intercept)	3·51	0·41	8·63	<0·001	4·07	0·68	5·96	<0·001
Status – *MPA*	−1·04	0·40	−2·62	0·046				
Technique – *OC*	0·54	0·22	2·53	0·012	0·92	0·47	1·94	0·066
Length	0·01	0·00	9·12	<0·001	0·01	0·00	3·89	<0·001
Group size	−0·83	0·31	−2·72	0·007	−1·30	0·17	−7·79	<0·001
Scarines								
(Intercept)	1·03	0·50	2·05	0·047	0·65	0·48	1·35	0·178
Status – *MPA*	−0·77	0·42	−1·84	0·128	−0·49	0·36	−1·37	0·179
Technique – *OC*	1·94	0·38	5·14	<0·001	0·33	0·28	1·18	0·241
Status*Technique	−1·00	0·43	−2·31	0·021	1·05	0·39	2·67	0·008
Length	0·01	0·00	7·08	<0·001	0·02	0·00	8·59	<0·001
Group size					−0·31	0·14	−2·25	0·025
Chaetodontids								
(Intercept)	4·16	0·27	15·46	<0·001	1·68	0·53	3·16	0·002
Length	−1·01	0·16	−6·20	<0·001				
Group size					0·02	0·00	3·50	0·001

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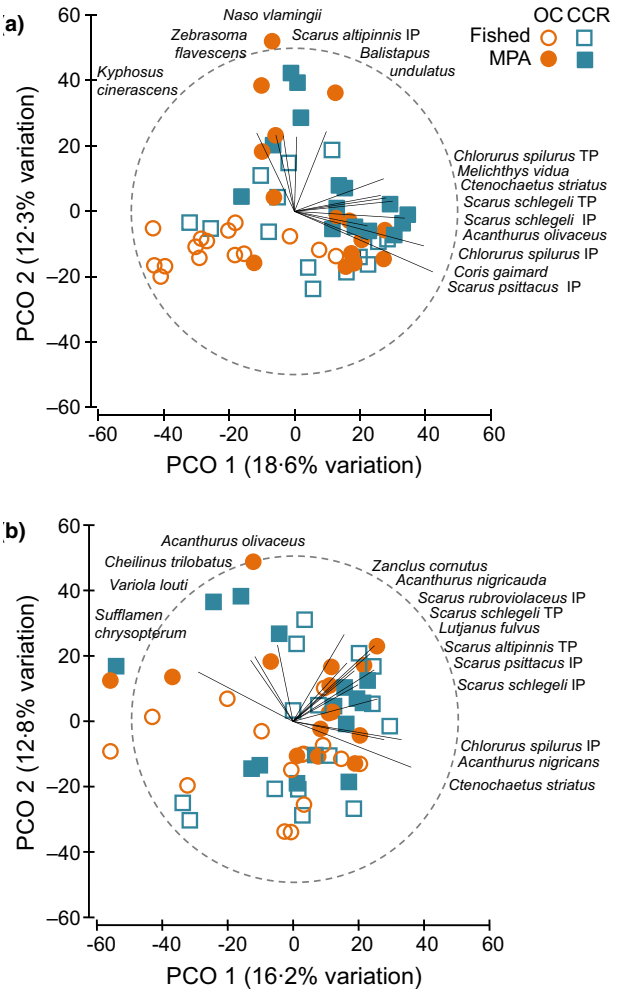

Fig. 3. Principal components analysis (PCO) for fish assemblage biomass at (a) Guam West and (b) Guam South. Species correlations are indicated by the length and direction of vectors.

Table 2. Number of high-value individuals recorded for each diving technique at MPA and fished areas. Data summed across locations

	Fished		MPA	
High-value species	OC	CCR	OC	CCR
Cephalopholis argus	–	–	–	2
Naso tonganus	–	–	–	10
Lethrinus rubrioperculatus	–	–	2	–
Cetoscarus bicolor	–	–	4	–
Caranx melampygus	–	–	2	1
Epinephelus fasciatus	–	–	2	2
Siganus punctatus	–	–	2	2
Variola louti	–	–	3	7
Scarus festivus	–	–	13	9
Carcharhinus melanopterus	–	1	–	–
Lethrinus harak	–	1	–	–
Acanthurus lineatus	–	7	–	–
Acanthurus triostegus	–	8	–	–
Siganus spinus	–	13	–	–
Lutjanus gibbus	–	1	–	3
Chlorurus microrhinos	–	4	–	2
Lethrinus obsoletus	–	5	–	1
Hipposcarus longiceps	–	1	1	–
Lutjanus monostigma	–	1	5	1
Siganus argenteus	–	2	2	2
Naso unicornis	–	2	2	8
Scarus rubroviolaceus	–	6	12	15
Lutjanus fulvus	–	6	5	7
Kyphosus cinerascens	–	10	41	27
Epinephelus merra	1	1	–	–
Macolor spp	1	2	–	2
Monotaxis grandoculis	1	2	10	27
Scarus altipinnis	2	6	58	24
Aphareus furca	3	1	12	7
Scarus schlegeli TP	7	10	12	17
Naso lituratus	22	56	68	74
Total	37	146	256	250

particularly the numerically dominate initial phase (IP) parrotfish *Chlorurus spilurus* and *Scarus psittacus*.

HIGH-VALUE SPECIES

Fifteen of the 31 high-value species were not observed using OC SCUBA in fished areas, but were recorded with CCR (Table 2). Yet there were no occurrences where high-value species were seen in fished areas with OC but not observed with CCR. Some species were not observed at all with OC, including the large, commercially valuable parrotfish *Chlorurus microrhinos,* two species of emperor *Lethrinus* spp. and the snapper *Lutjanus gibbus*. In fished areas, the total number of high-value fish counted was almost 4 times greater (37 vs. 146) when using CCR compared to OC, whereas numbers were similar (256 vs. 250) in protected areas.

Discussion

Our use of a silent diving system highlighted significant inaccuracies associated with the use of conventional SCUBA to collect visual census data on fish assemblages. These inaccuracies were most pronounced when surveys were conducted on reef fish that are targeted through spearfishing. A principal behavioural change in these exploited communities was strong diver avoidance, which resulted in depauperate assessments when using open-circuit (OC) SCUBA. Use of a silent diving system moderated this behaviour to the extent that species richness values were 48% higher and estimates of relative abundance and biomass ~200–300% greater within heavily fished areas. These results highlight the general importance of identifying all major sources of error associated with sampling populations in such variable environments. More specifically, it indicates how such errors can result in erroneous conclusions being drawn on the impacts of fishing and the effectiveness of MPAs. Given the ubiquitous nature of SCUBA assessments in marine research and the pre-eminent role of MPA's as management bulwarks against overfishing, these results have widespread implications and validity.

By producing depressed counts of fished species in unprotected areas, the use of OC SCUBA resulted in inflated assessments of the effectiveness of MPAs. Differences between OC and CCR were largely negated when

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counts were made within the MPAs where fish were not unduly disturbed by the presence of OC divers. In contrast, our control species, the butterflyfishes, which are not a target for fishing, showed no differences between protected and fished areas nor between diving techniques. It was suggested as long as fifteen years ago that the acquired behaviour of fish could result in erroneous conclusions when comparing marine reserves and fished areas using diver surveys (Kulbicki 1998). Our study expands on this by eliminating a major source of bias (the bubbles from SCUBA) and clearly demonstrates how changes in fish behaviour due to the presence of an observer using OC SCUBA can change the interpretation of the data collected.

Closed-circuit rebreathers are thought to minimize diver disturbance due to their silent operation underwater (Hanlon *et al.* 1982; Lobel 2001, 2005; Radford *et al.* 2005). A previous study by Cole *et al.* (2007) also performed quantitative fish surveys using different SCUBA diving systems to specifically investigate diver noise. However, in their case, noise was only reduced rather than eliminated, through the use of semi- rather than fully-closed-circuit rebreathers. They found few and largely inconsistent differences in fish abundance or size structure between diving techniques when sampling inside and outside marine reserves in New Zealand.

Fish are able to detect bubble sounds above that of rough sea conditions at distances of up to 200 m for OC SCUBA and up to 7 m for semi-closed-circuit rebreathers (Radford *et al.* 2005). This distance to detection remains substantial compared to a CCR which produces sound that should be virtually undetectable by fish at distances greater than 0·3 m from the diver (Radford *et al.* 2005). Apart from the differences in diving techniques, the larger observed effects in our study compared to that of Cole *et al.* (2007) could be due to increased diver avoidance behaviour of fishes in Guam, where spearfishing is frequently practiced. SCUBA can be used for spearfishing in both New Zealand (Cole *et al.* 2007) and Guam (Lindfield, McIlwain & Harvey 2014), and although use of SCUBA for spearfishing in Guam is predominately performed at night, this may increase diver avoidance compared to locations where SCUBA spearfishing is banned or less frequently practiced.

Sound travels considerable distances underwater, well in excess of the visible range of marine animals (Radford *et al.* 2005). The lower abundance, biomass and absence of many fishery-targeted species in surveys conducted with OC compared to CCR suggest that these species avoid divers outside of the range of visibility or transect dimensions. Although it is difficult to quantify fish reactions to a moving diver outside the range of visibility, the few studies that have used underwater sonar recorded fish actively avoiding divers (Chapman *et al.* 1974; Stanley & Wilson 1995; Schmidt & Gassner 2006). For example, Stanley & Wilson (1995) found that in the presence of SCUBA divers, fish would leave the area, reducing fish density by 60% on average. Furthermore, Chapman *et al.* (1974) used underwater speakers to mimic the noise of diver exhaust bubbles, confirming sound was the stimulus for fish reactions towards divers. As fish can detect divers outside the range of visibility or transect dimensions, diver surveys will ultimately underestimate true population size if fish respond negatively (Stanley & Wilson 1995; Kulbicki 1998; Bozec *et al.* 2011; Dickens *et al.* 2011).

Nevertheless, visual census remains the most popular method for assessing reef fish distribution and abundance. The rationale for this is that if fish behaviour towards divers is constant over the spatial and temporal scales investigated, then this should not impact the relative patterns and interpretation of data. However, our results suggest otherwise and support a growing body of evidence that fish behaviour towards divers is strongly related to the level of fishing pressure (Kulbicki 1998; Bozec *et al.* 2011; Januchowski-Hartley *et al.* 2011, 2013). Surveys using OC SCUBA recorded the relative abundance of fish 1·5–2·6 times greater in MPAs than fished areas, a result not mirrored with CCR surveys. A similar magnitude of difference was calculated by Kulbicki (1998) when using distance-based sampling to account for trends in fish detectability with increasing fishing pressure. Shy fish species were expected to differ in abundance by a factor of 1·8 between protected and fished areas if using 5-m-wide strip transects (Kulbicki 1998). As we found targeted fish species to respond negatively to OC divers at heavily fished locations, visual census will often underestimate fish abundance. When comparing fish populations across spatial scales, such changes in fish behaviour will result in overestimates of the difference between heavily fished and lightly or unfished reference areas. This is an important and widely encountered source of bias that should be acknowledged and minimized where possible.

Not only is the behaviour of fish towards divers likely to change between diving techniques and fishing pressure, but reactions will also be species specific (Kulbicki 1998; Watson & Harvey 2007). For example, Kulbicki (1998) classified a subset of reef fish as 'shy' species which tend to stay away from the observer. Of these high-value fishery-targeted species, large parrotfish *Chlorurus microrhinos*, *Scarus rubroviolaceus*, *Scarus altipinnis*, *Hipposcarus longiceps,* unicornfish *Naso unicornis* and the emperor *Lethrinus harak* were not observed in this study while using OC SCUBA in fished areas, but they were all recorded in the surveys conducted on CCR. The discrepancy in the numbers of high-value species recorded between diving techniques in this study is of particular concern for monitoring and management as they are important food resources on coral reefs (Houk *et al.* 2012). Closed-circuit rebreathers provide a means to reduce diver avoidance and therefore improve the accuracy of fish surveys.

The two MPA systems utilized in this study were located in very different usage zones on Guam, with the southerly site more remote and subject to less fishing and general visitor pressures. Our results for this location using the CCR system indicated no significant difference in the counts of targeted fish inside and outside the MPA. Contrast to this with the OC surveys which concluded that there were significant differences between fished and protected areas. These differences reflect the ability of the CCR surveys to pick up those larger, heavily

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targeted fish species that retreat from divers using OC. This result also favours a different interpretation of fishing effects in the south of Guam, namely that fished areas are still holding similar fish stocks to the MPA. Whether this reflects lower fishing pressure outside the MPA or evidence of poaching inside the MPA is a hypothesis for future study.

Although CCR diving has long been encouraged for behavioural observations of marine life (Hanlon *et al.* 1982; Lobel 2001, 2005) and for increased efficiency during deep technical diving (Parrish & Pyle 2002; Sieber & Pyle 2010), this study is the first to justify their use for quantitative fish surveys. While rebreather systems are more expensive than open-circuit SCUBA equipment and require additional training for use and regular maintenance, they are rapidly gaining popularity in the recreational diving industry with new improvements in design, logistical support and acceptance by dive training agencies. Despite this, the widespread use of CCR will be limited due to logistical and cost limitations and alternative methods from OC SCUBA-based surveys may warrant the simple use of snorkelling for assessing shallow depths (Taylor & McIlwain 2010) or using remote underwater video stations when sampling deeper waters (Mallet & Pelletier 2014). Overall, this research highlights the need for minimizing observer disturbance where possible and we recommend further studies to test the applicability of our results to other locations.

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Data accessibility

Data presented in this study can be accessed through the Dryad digital repository (Lindfield *et al.* 2014).

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Fig. S1. Map of sampling locations around Guam. Circles represent MPA sites and stars represent fished sites.

Table S1. Species list and groupings of all fish species that were counted during dive surveys.

Table S2. Summary table of the GLMM analysis examining the response of abundance, biomass and species richness of fished species between MPA status and diving techniques.

Table S3. Summary table of the GLMM analysis examining the response of acanthurid, scarine and chaetodontid abundance between MPA status and diving techniques.

Table S4. Three-way PERMANOVA testing for differences in fish assemblage structure at each location. *P*-values were obtained using permutation tests (9999 permutations) for each individual term in the model.

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